

Original LCTRSs

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times Loc \Rightarrow State \\ \text{noncrit1} : Loc \\ \text{crit1} : Loc \\ \text{noncrit2} : Loc \\ \text{crit2} : Loc \\ \text{unlock} : Loc \\ \text{lock} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{l} 1 : \text{state}(\text{noncrit1}, p2, \text{unlock}) \rightarrow \text{state}(\text{crit1}, p2, \text{lock}) \\ 2 : \text{state}(\text{crit1}, p2, \text{lock}) \rightarrow \text{state}(\text{noncrit1}, p2, \text{unlock}) \\ 3 : \text{state}(p1, \text{noncrit2}, \text{unlock}) \rightarrow \text{state}(p1, \text{crit2}, \text{lock}) \\ 4 : \text{state}(p1, \text{crit2}, \text{lock}) \rightarrow \text{state}(p1, \text{noncrit2}, \text{unlock}) \\ 5 : \text{state}(p1, p2, p3) \rightarrow \text{success} \\ 6 : \text{state}(\text{crit1}, \text{crit2}, p3) \rightarrow \text{error} \end{array} \right\}$$

APR Problem

$$\{ 7 : \text{state}(\text{noncrit1}, \text{noncrit2}, \text{unlock}) \Rightarrow \text{success} \}$$